



GENERAL GUIDE

Ndarama Grid User Guide

A practical guide to the desktop application, Android companion and site workflows



PitGuard CAM-02 image shown inside the application. Demonstration imagery, not a live camera feed.

MineTech Innovation Challenge Zimbabwe 2026

Prepared 30 September 2026

What Ndarama Grid does

Ndarama Grid is an offline-first mine-operations application. The Windows desktop app runs the local Edge Hub and presents 12 work areas. The Android app can work from its bundled sample data or connect to a PC Hub over a shared Wi-Fi network. A site computer stores the Hub data locally in SQLite.

Area	Purpose
Command Center	Summarize site status and open other windows.
PitGuard Safety	Review sensor fields, PPE status, incidents and alert controls.
Drone Map and Tag Manager	Review site plans, tag fleet status and recorded scans.
Mesh Overview and OreChain Ledger	Review message queues, parcel records, custody history and QR verification.
Copilot, Planning, Assets and Workforce	Support advisory questions, work planning, equipment and shift checks.
Mining News and Settings	Read mining updates and configure profile, site, language, data and sync.

Data labels matter

Look for the sync or source status in each window. A live connection, cached record and bundled demonstration record are different sources. Values shown in the default site are sample data and must not be treated as live mine measurements unless the site has connected and validated the intended hardware.

Start the desktop application

- On Windows, start Ndarama Grid with `launch_gui.bat` or open the current desktop build.
- Wait for the dashboard to load. If the Hub does not start, check that the application is allowed by Windows Firewall on a private network.
- Use the left navigation to open a window. Choose the active site from the site selector at the top of the application.
- Open Settings to check the operator profile, active site, roles, language and Hub connection details.

The dashboard

The Command Center summarizes PitGuard, OreChain, Mesh and Copilot status, then shows a site watchlist, mineral mix, planning outlook and indicative commodity board. Select a summary card or navigation item to open its detailed window. The dashboard's sample values are illustrative until connected to a configured site data source.

Use the status indicators

- Live or synced indicates that the view has received data from the Hub.
- Cached indicates that the view is using a previously stored response.
- Demo or sample values support interface evaluation only.
- Offline means the Hub cannot currently be reached; cached and bundled data may remain visible.

Connect Android to the PC Hub

The Android app cannot use the phone's 127.0.0.1 address to reach the PC. Use the PC's LAN address and the Hub port shown by the desktop application.

Step	Action
1	Start the desktop application on the PC and keep it running.
2	Open PC Settings, then Data & Sync. Copy one of the displayed Hub addresses.
3	Connect the phone and PC to the same trusted Wi-Fi network.
4	On Android, open Settings > Data & Sync, enter the full PC address and any configured bearer token, then select Save address.
5	Select Test connection. A successful response confirms that Android can reach the PC Hub.

The desktop launcher usually selects port 8000. If another service occupies it, the launcher can choose a nearby port; always use the full address shown in PC Settings. If the connection fails, check Wi-Fi isolation, Windows Firewall, the selected address and whether the PC app remains open.

Data flow

After connection, Android requests and submits Edge Hub records over the local network. The Hub remains on the PC. Camera QR verification sends the decoded code to the Hub, where it is looked up against the stored parcel ledger. The Android camera image stays on the phone.

Review PitGuard and safety information



PitGuard CAM-02 underground-drive imagery shown in the application. This is interface imagery, not a live feed by itself.

Open PitGuard Safety to view available methane, carbon monoxide, oxygen, seismic, dust, PPE, tailings and incident information. The exact fields shown depend on the connected site data. Check timestamp, source and device status before relying on a measurement.

- Use calibration controls only under the site's approved sensor procedure.
- Escalate an alert through the mine's established emergency plan. Confirm actual alarm devices and communication channels independently.
- Treat PPE computer-vision flags as review prompts, not final decisions about an individual's compliance.
- Do not use dashboard values as a substitute for certified safety instruments or trained supervision.

PitGuard controls and alert cards may be demonstrations until their hardware and communication links have been configured and validated. The app itself does not certify compliance with SI 109 or any site safety plan.

Record and verify OreChain parcels

OreChain stores parcel details and custody events. In the Ledger window, select a parcel to review its mineral, weight, status, holder and custody history. Hash-chain and signature indicators describe the records the Hub can verify; they do not prove the physical mineral matches the record.

Scan a parcel QR code on Android

- Open OreChain Ledger and select Scan code. Grant camera permission when Android prompts.
- Hold the QR code in the camera view until the parcel value is read. The image stays on the phone.
- The app sends the decoded code to the configured PC Edge Hub for a parcel lookup and chain check.
- Read the result carefully: signature verified, chain intact, missing parcel and failed integrity check have different meanings.
- If camera scanning is unavailable, type the printed QR value or parcel ID and select Verify parcel authenticity.

A successful digital integrity check confirms that the stored record matches its chain and signature checks. It does not independently authenticate physical contents, legal title or counterparty identity. Use the receiving site's inspection and transaction procedures as well.

Work offline and troubleshoot

The desktop Hub stores data locally. The interface may show cached or bundled sample data when the Hub, network or external service is unavailable. Mesh Overview presents a store-and-forward queue concept, including higher priority for safety messages. Real radio behavior depends on configured and tested network hardware.

Common checks

Symptom	Check
Android cannot connect	Use the exact PC address and port shown in PC Settings. Confirm same Wi-Fi, PC app running, private-network firewall access and no guest-network isolation.
401 or token rejected	Confirm the bearer token configured for the PC Hub. Save it in Android Settings and test again.
QR code returns no parcel	Check that the scanned code belongs to a parcel in this PC Hub's ledger. Verify that Android is connected to the correct PC and site.
Camera does not open	Allow camera access for Ndarama Grid in Android settings. Use manual QR entry if scanning remains unavailable.
Images do not load	Update to the current Android APK. The image files are packaged with the app; retry after reinstalling the supplied build.
Data appears old or illustrative	Check the source badge, sync time, active site and device connection. Do not interpret sample records as operational readings.

Operational limits

The available project documentation describes prototype software and conceptual hardware. No claim of regulatory approval, certified safety performance, validated radio coverage, live customer deployment or independently audited ledger workflow is made by this guide. Site owners should complete engineering, safety, privacy, security and legal review before production use.